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Pre-ictal heart rate changes: a systematic review and meta-analysis

Elisa Bruno^{a,b}, Andrea Biondi^{a,b}, Mark P Richardson^{a,b}, on behalf of the RADAR-CNS Consortium^b

^a Institute of Psychiatry, Psychology & Neuroscience, Division of Neuroscience, King's College London, UK

^b www.radar-cns.org

Corresponding author

Professor Mark P Richardson

Institute of Psychiatry, Psychology & Neuroscience

Division of Neuroscience

King's College London

Maurice Wohl Clinical Neuroscience Institute

Ground Floor (G.39), 5 Cutcombe Road, Camberwell, London, SE5 9RX

Tel: (+44) 20 7848 5429

mark.richardson@kcl.ac.uk

Introduction

The alterations provoked by epileptic seizures on cardiac rate and rhythm have been recognized for more than 70 years.^{1,2} Ictal cardiac changes have been attributed either to the activation of cortical structures connected to the autonomic centres, or to peripheral mechanisms regulating reflex responses, driving various cardiac manifestations.³ Although a reduction of the heart rate (HR) can occur, the most commonly observed pattern associated with seizures is represented by an increased HR.³ Ictal tachycardia (IT) has been reported in up to 80% of all seizures and in 82% of people with epilepsy on average,⁴ while ictal bradycardia (IB) is commonly considered a rare event, manifesting in less than 5% of epilepsy patients.^{5,6} The possibility to detect alterations of cardiac parameters during a seizure has led to consider the HR as a potential extra-cerebral seizure biomarker.⁴ In comparison to other more invasive measurements, HR assessment has the potential of providing a reliable and easily measurable signal which may be recorded continuously in a daily life environment. In addition, HR alterations have been reported to occur early during a seizure or even pre-ictally,⁷ demonstrating a theoretical role in seizure prediction. However, studies investigating ictal and pre-ictal HR changes have yielded controversial results and our understanding of the mechanism supporting these alterations is still limited.³ In fact, the frequency of ictal HR manifestations in patients with epilepsy varies considerably across studies and estimates range between 38-100% for IT and <5-66.7% for IB,^{3,4} with higher variability when a pre-ictal onset is considered. This variance likely reflects differences in measurements along with clinical variables, including epilepsy and seizure type, age groups, use of antiepileptic drugs (AEDs), lobe of seizure onset and presence of cardiac co-morbidities. In addition, methodological issues can also be responsible for the mixed findings, including methods used to assess seizure onset and to define HR changes. Understanding the distribution of pre-ictal HR alterations and the temporal relation between these changes and seizure activity might represent a pivotal point to shed new light on the mechanisms shaping the ictal autonomic changes and influencing seizure spreading patterns. Moreover, this could also help to explore the utility of this measure as a seizure detection tool and to identify the target population who might benefit from its use. We aimed to perform a meta-analysis to assess the incidence of pre-ictal HR alterations and to identify clinical and study-related factors modulating this phenomenon.

Methods

This systematic review and meta-analysis was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and Cochrane Handbook guidelines.^{8,9}

Criteria for considering studies for this review

Retrospective and prospective studies (case-control, cohort or case series) recording concurrent pre-ictal EEG and HR in adults and children with epilepsy were considered for inclusion. Studies focusing on neonatal seizures only were excluded. Pre-ictal HR changes were classified as HR reduction (HRR) or increase (HRI) as compared to the baseline HR. Methods used to measure the HR and definitions of HR alterations were extracted and considered as separate variables assessed in the meta-regression as potential sources of between studies heterogeneity.

Search methods for identification of studies

A systematic search with no language restrictions was carried-out to identify all relevant published and unpublished studies. The search strategy included the terms “epilepsy”, “heart rate”, “ictal tachycardia” and “ictal bradycardia” and is reported in Appendix S1. The search was conducted from the first date available (1958) up to May 30, 2017 in MEDLINE, Embase and the Cochrane Central Register of Controlled Trials (CENTRAL). The search strategies for each database were based on the strategy developed for MEDLINE, taking into account the differences in controlled vocabulary and syntax rules. In addition to the electronic searches, we hand-searched reference lists of all available review articles and primary studies and hand-searched the references quoted in the most recent congress proceedings (e.g. International Epilepsy Congress, European Congress on Epileptology).

Data collection and analysis

Two review authors (EB and AB) independently assessed the titles and abstracts of all the studies identified by the electronic searching or hand-searching. Full texts of potentially relevant studies were obtained and screened. We resolved any disagreements concerning study inclusion and exclusion by discussion. For each study included, two review authors (EB and AB) independently extracted the following data on an ad-hoc created data collection form: study design (prospective, retrospective) and setting (inpatients, outpatients); demographic and clinical data of the population (number of patients, age group, gender, type of epilepsy, drug-resistance, cardiac comorbidities, AEDs administration/withdrawal during the recording); number of seizures recorded, seizure type and focus (including focus side), definition of seizure onset (EEG, clinical, either EEG or clinical); EEG characteristics: type (scalp, intracranial), number of electrodes used (standard, non-standard), duration (continuous, intermittent, number of hours recorded), use of video recording; HR assessment methods (automatic versus manual count of R-R intervals or QRS complexes or beats) and length of ECG epoch used for HR analysis (in seconds); duration of the pre-ictal period assessed (in seconds); number of seizures presenting with ictal HR change (IT, IB and definitions adopted) and ictal HR peak/minimum reached; number of seizures with pre-ictal HR change (pre-ictal HRI and HRR) and onset time (seconds before seizure onset).

Quality assessment

The quality of included studies was evaluated using a standard assessment tool, that was slightly re-adapted (Appendix S2), and included sample representativeness, condition assessment, and statistical methods.¹⁰ Each study was given a quality score of 0 to 8 based on fulfilment of the quality criteria. The quality score was considered as a separate variable in the meta-regression for studies included in the meta-analysis.

Data synthesis and analysis

Studies reporting the total number of seizures recorded and the number of seizures with pre-ictal HR changes (including 0), were included in a meta-analysis. **Two reviewers independently assessed the risk of bias in included studies considering the following domains: definition of HR change, definition of pre-ictal period, time interval used to calculate the HR change, assessment of onset of HR change. The overall 'Risk of bias' assessment was based on recommendations in the Cochrane Handbook for Systematic Reviews.⁹ If we assessed one or more domains as being at high risk of bias, we rated the overall score as high. If we rated all domains as being at low risk of bias, we considered the overall score to be low. We rated studies with all the other combinations as being at unclear risk of bias overall.**

Pre-ictal HRI and pre-ictal HRR were separately analysed. In addition, meta-analysis was separately fitted according to epilepsy type, age group and AEDs administration/withdrawal, when these variables were specified by an adequate number of studies. We used the 'metaprop_one' command in Stata 14.0 to estimate crude incidence rates along with their 95% confidence intervals (CI) and we expressed the estimates as the number of seizures with pre-ictal HR changes per 100 seizures. We reported the pooled, weighted estimate generated by random-effects models. To handle the studies with zero events, we used Freeman–Tukey double arcsine transformation which stabilizes the variance of the proportion restricting the 95% CI within the range of 0 and 1, even in the presence of zero events.¹¹ As a sensitivity analysis, the pooling process was repeated after the successive removal of incidence studies with a low-quality score. The I^2 was used to quantify the magnitude of between-study heterogeneity and the Cochrane Q statistic was calculated to determine significance. Publication bias was investigated statistically using Begg's and Egger's tests. To determine the influence of the clinical variables and of the study-level factors on the observed variability, we used random-effects meta-regression. We regressed one variable at a time. Significance level was established at $p < 0.05$. All analyses were performed using STATA version 14.0 (StataCorp, College Station, TX, U.S.A.).

Results

Study selection and quality assessment

The search of electronic databases yielded 1,130 references (Figure 1). One additional study was identified by hand-searching. After duplicates and non-relevant studies were removed, the titles and abstracts of the remaining studies were reviewed and the full-text of 98 articles with potentially relevant studies was assessed. Finally, 30 published studies were considered eligible for qualitative synthesis. All the included studies assessed the occurrence of pre-ictal HRI, while pre-ictal HRR was investigated in 20 studies only. The characteristics of all the 30 studies included are summarized in Table 1 and more extensively detailed in Table S1 (Supporting Information). Quality assessment revealed a median study quality score of 6/8 (range 1-8).

Pre-ictal HRI

Summary of characteristics of the studies

Thirty studies, enrolling 1,110 people living with epilepsy (39.0% female; 38.2% male; 22.8% not reported) and a total of 2,957 seizures, assessed the occurrence of pre-ictal HRI. Eighteen included adults,¹²⁻²⁹ four children only,²⁹⁻³³ and seven both adults and children.³⁴⁻⁴⁰ The age group was not reported in one study.⁴¹ Fourteen studies included people with different type of focal epilepsy,^{12-20, 34-37,41} 11 TLE,²¹⁻³¹ four a combined sample of focal and generalized epilepsy,^{32,233,38,39} and one nocturnal frontal lobe epilepsy (NFLE).⁴⁰ Fifteen studies included drug-resistant patients,^{14-17,20-22,26,28,30,33,35} three non-refractory patients and 12 did not report if patients were drug-resistant or not.^{12,13,24,25,27,29,31,32,34,36,38,41} In ten studies people continued to receive their AEDs during the recording,^{15,16,20-22,26,30,31,34,39} while treatment was withdrawn in six studies,^{21,25,27,29,32,40} and 14 did not report the AEDs status. Twenty-one studies included complex partial seizures (CPS) alone or in combination with other seizure types.^{15-23,26-31,33,35-40} All the studies used scalp EEG, with five studies including also intracranial EEG.^{12,19,24,25,28} The majority were based on prolonged EEG recording, using a standard number of electrodes and additional video recording. Only studies published before 1997 used less electrodes than standard.^{16,17,29,33,34} HR assessment was specified in 23 studies, and it was either manual (nine studies)^{21,24,25,30,34-38} or automatic (14 studies).^{12,14-17,20,22,23,26,29,33,39-41} The mean time-epoch considered for HR assessment was on average 23.1 seconds (± 23.3 ; range 5-60) and a mean pre-ictal period of 99.2 seconds prior to seizure onset (± 156.5 ; range 15-600) was assessed. One study considered multiple pre-ictal periods, starting from 240 minutes before seizures onset.¹⁷

Frequency of ictal and pre-ictal HRI

The mean baseline HR was 74.35 bpm (± 4.8 ; range 65.9-86.4). Although all the studies described the occurrence of IT, only 18 studies clearly reported its frequency, giving a cumulative incidence of 1,556/2,957 seizures (52.6%).^{12-15,20,22,23,26,29,30,32-39} The remaining eight reported the occurrence of IT without specifying the frequency,^{18,19,21,24,25,27,28,30} while four did not report any quantitative data.^{16,17,31,41} The mean ictal HR peak was 116.5 bpm (± 27.0 ; range 81.2-201.0), demonstrating a

mean increase of 56.8% as compared to the baseline HR. Pre-ictal HRI was found in 623/2,957 seizures (21.1%) and it was reported in 14 studies,^{12,14,21,22,24-26,28-33,35,39} while no pre-ictal change was found in five studies.^{16,19,27,28,34} The remaining eleven studies did not report the number of seizures with pre-ictal HRI, although they commented on its occurrence and onset time.^{13,15,17,18,20,23,36-38,40,41} The mean onset time of pre-ictal HRI was 70.4 seconds prior to seizure onset (± 128.9 ; range 1.0-420.0), with one study reporting HR increase from 30 minutes before.¹⁷

Clinical variables and pre-ictal HRI

In five studies,^{14,19,26,29,30} seizures originating from the mesial temporal lobe were more often associated with pre-ictal HRI as compared to lateral and extra-temporal onset, with two studies reporting more significant pre-ictal HRI in right-sided mesial seizures.^{18,29} Only one study observed that pre-ictal changes were more frequent in patients with lateral TLE.¹² One study investigating NFLE reported a pre-ictal HR increase, although physiological arousal was also associated with a similar HR alteration.⁴⁰ Seizures of frontal origin were also associated with a lower pre-ictal HR compared to temporal lobe seizures.¹⁵

Seven studies included focal seizures exclusively and six,^{20,21-23,26,29} reported the occurrence of pre-ictal HRI while one did not observe any alterations.²⁷ Patients with secondarily generalized tonic-clonic (SGTC) seizures were associated with increased pre-ictal HR in one study compared to those with localized seizures only.¹⁵ However, this finding was not confirmed in another study using a paired design methodology.²⁸ Although male gender was more often associated with HRI,^{18,25,26,36} no data were available on the influence of gender on HRI onset time. The peri-ictal HR pattern was considered consistent or predictable in the same patient across multiple seizures in 2 studies,^{14,22} while no stereotyped HR behaviour and high intra-individual variability was observed in another study.³⁴

Meta-analysis

Fifteen studies analysing pre-ictal HRI were considered eligible for the meta-analysis.^{12,14,21,22,24-27,29-33,35,39} For studies included in the meta-analysis, the quality assessment yielded higher scores (7/8). The pooled incidence of pre-ictal HRI was 36/100 seizures (95% CI 22-50) (Figure 2A), with a significant between study heterogeneity ($I^2 = 97.25\%$, Q p-value=0.000). Egger's and Begg's test were not significant ($p = 0.74$, $p = 0.92$ respectively) indicating a low risk of publication bias. The analysis performed according to epilepsy type (Figure 3A) showed that studies including TLE had a pre-ictal HRI incidence of 44/100 seizures (95% CI 33-55) while the incidence was 36/100 seizures (95% CI 6-75) for studies including mixed focal types and 16/100 seizures (95% CI 4-34) for those including generalized and focal epilepsy combined. Studies including adults presented with an incidence rate of 55/100 seizures (95% CI 41-68) while the incidence was 25/100 seizures (95% CI 9-44) for studies

including children and 9/100 seizures (95% CI 0-33) for studies including both children and adults (Figure 3B). Studies including patients recorded while on AED treatment had a pooled incidence rate of 35/100 seizures (95% CI 16-58) as compared to 22/100 seizures (95% CI 4-47) for those studies including patients withdrawing the treatment (Figure 3C). The treatment status was not reported in 5 studies.^{12,14,24,33,35}

The meta-regression (Table 2) showed that, among clinical variables, the age group of study participants had a significant impact on variability of studies ($p=0.004$), accounting for 23.44% of the observed heterogeneity. Among methodological covariates, the length of the pre-ictal period during which the HR was assessed ($p=0.016$), explained an additional 33.44% of the observed heterogeneity, with higher incidence in studies using a longer time frame (60 seconds or more). In addition, the pre-ictal HRI estimates were significantly correlated with the IT frequency ($p=0.004$) and with the time of onset of the pre-ictal changes ($p=0.04$), and higher pre-ictal estimates were reported in studies observing higher IT estimates or earlier HRI presentations (Figure 4). Finally, the sensitivity analysis demonstrated comparable results when a low-quality study was excluded (pre-ictal HRI incidence 33/100 seizures, 95% CI 19-46; $I^2=97.2\%$, Q p -value=0.000) and the meta-regression found no effect of study quality on the estimates of the pre-ictal HR ($p=0.2$).

Pre-ictal HRR

Twenty studies investigated the presence of pre-ictal HRR. IB was reported in 88/2,346 seizures (3.7%) and was observed in 13 studies.^{12-15,20,21,26,30,32,34-36,39} One study reported pre-ictal HR reduction without commenting about subsequent IB,³¹ and six studies did not observe any IB.^{22,25,27,29,37,38} The mean baseline HR was 76.2 bpm (± 4.9 ; range 69.5-86.4) and the mean ictal HR minimum was 62.2 bpm (± 13.4 ; range 50.0-76.5), demonstrating a mean decrease of 18.4% as compared to the baseline HR. Pre-ictal HRR was observed in 13/2,346 seizures (0.5%) and was reported in four studies, exclusively including a paediatric population.^{15,30-32} Two studies included patients with refractory TLE,^{30,31} one reporting a mild pre-ictal bradycardia progressing into IT in seizure originating from the lateral temporal areas or from the right hemisphere.³⁰ The other two studies included a mixed sample of focal and generalized epilepsy,^{15,32} one describing changes in left focal seizures only.¹⁵ Time of onset of the pre-ictal changes was not reported.

Meta-analysis

Twelve studies analysing pre-ictal HRR were considered eligible for the meta-analysis.^{12,14,21,25,26,29-34,39} The pooled incidence of pre-ictal HRR was 0/100 seizures (95% CI 0-1) (Figure 2B), with a significant between study heterogeneity ($I^2=49.26\%$, Q p -value=0.03). Egger's and Begg's test were not significant ($p=0.99$, $p=0.11$ respectively) indicating a low risk of publication bias.

Discussion

This systematic review and meta-analysis of the incidence of pre-ictal HR changes demonstrated a pooled pre-ictal HRI incidence of 36/100 seizures, with HR manifestations preceding the clinical or EEG signs of 70.4 seconds on average. Higher estimates were found for studies including patients with TLE (44/100 seizures), adults (55/100 seizures) and people on AEDs (35/100 seizures). Using data reported in the literature, we identified an incidence of 0.5/100 seizures of pre-ictal HRR, reported exclusively in the paediatric population and a pooled incidence of 0/100 seizures through the meta-analysis.

To ensure a comprehensive and unbiased study selection, we decided a-priori not to apply any study inclusion restriction based on IT and IB definitions and favoured the terms pre-ictal HRI and HRR. In fact, at their onset, the HR alterations may occur gradually and may not satisfy the current definitions. In addition, the HR threshold to define IT and IB is not constant and may be influenced by various factors, including age, gender and other clinical variables, severely limiting the applicability of a single definition.

Although the mechanisms supporting cardiovascular changes in epilepsy are largely unknown, there is evidence of the involvement of cortical structures, constituting the central autonomic network (CAN, including the amygdala and the insular cortex), and of adaptive cardiovascular reflexes triggered by ictal manifestations, especially motor.⁴² It has been previously suggested that the variable ictal cardiac alterations reported across studies (IT, IB, combined IT/IB and oscillatory HR patterns) may reflect the different contribution of the central and peripheral autonomic pathways driving distinct cardiac manifestations.³ To attenuate the interference of the peripheral adaptation responses on study results, we limited our analysis to those HR manifestations occurring pre-ictally and thus, more likely consequent to CAN involvement.

The majority of the articles reviewed evaluated seizures originating from the temporal lobe and we observed a high incidence of pre-ictal HRI in those specifically including TLE patients.

Temporal lobe seizures have been more often associated with ictal HR alterations as compared to seizures of extra-temporal origin.³ Moreover, autonomic features conventionally tend to precede other clinical presentations in temporal seizures and this might well apply to HR manifestations, supporting our findings.⁴³ It appears plausible that the anatomic location of the temporal lobes facilitates the propagation of seizures to the CAN with consequent cardiovascular activation.⁴⁴ To corroborate this hypothesis, we found that pre-ictal HR manifestations were more frequently reported in mesial TLE, involving areas sitting closely to the CAN centres, as compared to lateral TLE or other lobe origin. However, the higher pre-ictal HRI incidence in mesial temporal lobe seizures

needs to be additionally verified by studies using the ictal EEG pattern to classify mesial and lateral seizures rather than clinical sign or MRI findings as often reported in the literature.

Current data did not provide enough evidence to prove a hemispheric lateralization of autonomic cardiovascular control, or a presumed right dominance in the sympathetic network.⁴⁴ In fact, seizure lateralization was infrequently reported in the studies analysed and was rarely used to assess the time of onset of early cardiac manifestations, which may be valuable as reported in some studies.^{18,29} Similarly, the effects of seizure type and of gender on the occurrence of pre-ictal cardiac manifestations remain unclear as well as the intra- and inter-individual consistency of presentations. The meta-analysis additionally showed a doubled incidence of pre-ictal HRI in studies focused on adults (55/100 seizures) as compared with those enrolling children only (25/100 seizures) and the age of study subjects was a study-level source of heterogeneity identified through the meta-regression. Conversely, pre-ictal HRR was observed in paediatric studies exclusively. These findings appear in line with the ictal vagal tone reduction and absent sympathetic activation observed in paediatric patients.⁴⁵ In addition, children higher parasympathetic tone and its progressive reduction with increasing age,⁴⁶ might explain the more prevalent IB in younger ages and the higher incidence of IT in adults.⁴⁷ However, additional studies assessing how the age affect the occurrence of seizure-related autonomic responses are needed. This might enrich our incomplete understanding of those central mechanisms leading to specific cardiac manifestations, while providing additional insight on other age-dependent autonomic dysregulation-supported phenomena, particularly sudden unexpected death in epilepsy (SUDEP). Interestingly, SUDEP occurs more commonly among adults (4/1,000)⁴⁸ as compared with children (0.2/1,000)⁴⁹ and more often in patients with highly refractory epilepsy, who are largely represented in the population included in this review, making our findings particularly relevant.

Studies including patients recorded while on AEDs presented a slightly higher incidence of HRI (35/100 seizures), as compared to those withdrawing their therapy during the recording (22/100 seizures). Administration of AEDs can modulate the cardiovascular manifestations in epilepsy and can potentially diminish or amplify the peri-ictal cardiac changes. Channel blocking agents have been associated with altered autonomic functions and higher vulnerability to arrhythmias during seizures.³ Moreover, baseline HR and ictal HR are higher in patients on several AEDs and higher risk for SUDEP has been reported in patients on polytherapy as compared to monotherapy.^{3,49} Whereas the indirect effect of AEDs on the autonomic nervous system has been previously investigated, our findings suggest a potential influence of AEDs on the CAN. However, a possible confounding effect of other clinical variables cannot be ruled out, as individual patient data (including, duration of epilepsy, seizure severity/frequency, AEDs type, dose, duration of the therapy, extent of AEDs

withdrawal etc.) were not available and as many studies did not report the treatment status, and additional investigations should look at the specific effect of AEDs in the modulation of CAN responses.

Among the methodological factors investigated as potential sources of between studies variability, the duration of the pre-ictal time-period used for the HRI assessment accounted for a significant proportion of study heterogeneity (33.44%), with longer time-windows producing higher estimates. This may reflect the early onset of the HRI, additionally supported by our finding of a mean pre-ictal HRI onset of 70.4 seconds prior to seizure onset, possibly missed by studies using shorted periods of observation. Interestingly, studies reporting higher IT estimates or earlier HRI onset resulted in higher pre-ictal HRI frequencies, likely reflecting methodological choices and more focused study aims which cannot be further clarified with our meta-analytic approach.

Conclusions and recommendations

Whether the pre-ictal cardiac changes observed in the reviewed studies are a direct effect of the CAN activation, secondary to reflexes induced by seizures with undetected onset, or rather the result of more complex sympatho-parasympathetic ictal mechanisms, remains unclear.

Nevertheless, despite their primary or secondary origin, our review demonstrated that pre-ictal HRI seem to occur many seconds prior to seizure onset and more frequently in specific groups of patients with epilepsy, namely TLE, adults and patients on AEDs. In these groups, monitoring of HR changes could be attempted to early identify seizures prior to the assumed clinical or EEG manifestations, opening the possibility of early interventions. However, our study demonstrated that additional effort is necessary to more clearly delineate those patients who might benefit from HR monitoring. Future studies on clinically well-defined populations should look more carefully at the time of onset of the HR changes and should define seizures onset according to their EEG patterns. Other potential sources of variability worth further exploring, including gender, wakefulness/sleep status and disease duration. Moreover, pre-ictal HRR assessment deserves additional attention as a rarer, poorly evaluated event.

To conclude, HR changes may potentially represent the earliest clinical change experienced by the patient and if the involvement of central networks is experimentally proved, those currently defined pre-ictal manifestations might instead be regarded as extremely valuable ictal or early ictal signs.

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Figure legends

Figure 1. Flowchart of the study selection process. HRI: heart rate increase; HRR: heart rate reduction.

Figure 2. Cumulative incidence of pre-ictal heart rate increase (HRI) (A) and pre-ictal heart rate reduction (HRR) (B)

Figure 3. Cumulative incidence of pre-ictal heart rate increase (HRI) by epilepsy type (A), age group (B) and AEDs status during the recording (C)

Figure 4. Meta-regression bubble plot showing the variation of the incidence of pre-ictal heart rate increase (HRI) regressed against the (A) proportion of seizures with ictal tachycardia (IT) and the (B) time of onset of pre-ictal HRI reported in the studies included.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. MEDLINE search strategy

Appendix S2. Quality Assessment Tool

Table S1. Characteristics of the 30 studies assessing pre-ictal HR changes

Tables

Table 1. Characteristics of the included studies (N=30)

Studies	No. of Participants	Age group	Epilepsy type	No. of seizures	Seizures with pre-ictal HRI (N, %)	Seizures with pre-ictal HRR (N, %)
Van Buren et al. 1958 ²¹	13	Adults	TLE	20	19 (95.0)	None
Blumhardt et al. 1986 ²²	26	Adults	TLE	74	35 (47.3)	None
Keilson et al. 1989 ³⁸	45	Adults and children	Focal and Gen	106	NR	None*
Galimberti et al. 1996 ³⁴	50	Adults and children	Focal	90	None	None
Vaughn et al. 1996 ³⁹	12	Adults and children	Focal and Gen	44	6 (13.6)	None
Novak et al. 1999 ²³	12	Adults	TLE	NR	NR	NA
Scherthanner et al. 1999 ¹²	41	Adults	Focal	127	86 (67.7)	None
Saleh et al. 2000 ²⁴	27	Adults	TLE	26	7 (26.9)	NA
Wilder-Smith et al. 2001 ¹³	37	Adults	Focal	37	NR	None*
Kirchner et al. 2002 ²⁵	21	Adults	TLE	21	6 (13.6)	None
Zijlmans et al. 2002 ³⁵	81	Adults and children	Focal	281	65 (23.1)	NR
Leutmezer et al. 2003 ¹⁴	58	Adults	Focal	145	110 (75.9)	None
Di Gennaro et al. 2004 ²⁶	68	Adults	TLE	257	134 (52.1)	None
Mayer et al. 2004 ³⁰	20	Children	TLE	72	20 (27.8)	3 (4.2%)
De Oliveira et al. 2007 ²⁷	27	Adults	TLE	59	None*	None*
Standridge et al. 2010 ³²	58	Children	Focal and Gen	244	18 (7.4)	6 (2.4%)
Nilsen et al. 2010 ¹⁵	38	Adults	Focal	38	NR	NA
Toth et al. 2010 ¹⁶	31	Adults	Focal	31	None*	NA
Surges et al. 2010 ²⁸	25	Adults	TLE	50	None*	NA
Calandra-Buonaura et al. 2012 ⁴⁰	10	Adults+1 child	NFLE	45	NR	NA
Behbahani et al. 2013 ¹⁷	12	Adults	Focal	133	NR	NA
Jansen et al. 2013 ³³	35	Children	Focal and Gen	80	25 (31.3)	3 (3.75%)
Chen et al. 2014 ³⁶	81	Adults and children	Focal	181	NR	NR
Kato et al. 2014 ²⁹	21	Adults	TLE	77	27 (35.1)	None
Behbahani et al. 2016 ¹⁸	24	Adults	Focal	232	NR	NA
Chen et al. 2016 ³⁷	102	Adults and children	Focal	202	NR	None*
Hampel et al. 2016 ¹⁹	37	Adults	Focal	45	None*	NA
Jaychandran et al. 2016 ²⁰	42	Adults	Focal	57	NR	NR
Moridani et al. 2017 ⁴¹	7	NR	Focal	11	NR	NA
Whealy et al. 2017 ³¹	49	Children and teens	TLE	172	65 (38.0)	1 (1%)

Gen: generalised; HRI: heart rate increase; HRR: heart rate reduction; NA: not assessed; NFLE: nocturnal frontal lobe epilepsy; NR: not reported; TLE: temporal lobe epilepsy

*cumulative data only (number and % of seizures not reported)

Table 2. Meta-regression of incidence of pre-ictal HR increase, univariate analysis

Covariates	No. of studies	p-value	Heterogeneity (τ^2)	Percent heterogeneity
Study design				
Prospective	2			
Retrospective	13	0.97	0.070	-7.52
Setting				
Inpatients	13			
Outpatients	2	0.42	0.067	-1.80
Number of participants				
≤35 (median)	8			
>35	7	0.96	0.071	-7.76
Age group				
Adults	8			
Children	4	0.004	0.034	47.08
Adults and Children	3			
Epilepsy type				
Focal and Generalized	3			
Focal	4	0.17	0.060	8.08
Temporal lobe epilepsy	8			
Drug-resistance				
Non-refractory	1			
Refractory	6			
Refractory and not	2	0.4	0.067	-1.56
Unclear	6			
Anti-epileptic drugs				
Off	3			
On	7	0.4	0.074	-3.57
Unclear	5			
Cardiac comorbidities				
None	8			
Not-assessed	7	0.36	0.066	-1.08
Number of seizure				
≤80 (median)	8			
>80	7	0.95	0.071	-7.76
Seizure type				
Complex partial seizures	5			
Multiple seizure types	4	0.89	0.071	-7.65
Not reported	6			
Epileptogenic focus				
Mainly right sided	6			
Mainly left-sided	2	0.51	0.068	-4.17
Right and left-sided	1			
Not reported	6			
Seizures presentation				
Clinical	3			
Subclinical	2	0.46	0.068	-3.36
Not reported	10			
Seizure onset definition				
EEG	8			
Clinical	2	0.42	0.067	-2.02
Either EEG or clinical	3			
Unclear	2			
EEG type				
Scalp	12			
Intracranial	2	0.49	0.068	-3.39
Scalp and intracranial	1			
EEG duration				
Continuous	13			
Intermittent	1	0.51	0.068	-3.37
Unclear	1			
No. EEG electrodes				
Standard	8			
Non-standard	4	0.76	0.070	-6.66
Unclear	3			
Video EEG				
Yes	10			
No	3	0.34	0.065	0.22
Unclear	2			
Heart rate assessment methods				
Automatic	7			
Manual	8	0.26	0.064	2.89
Pre-ictal time duration				
<60 sec (median)	6			
≥60 sec	7	0.016	0.043	33.44
Not reported	2			
ECG time-epoch				
<15 sec (median)	6			
≥15 sec	6	0.69	0.070	-6.21
Not reported	3			

Definition of ictal tachycardia and bradycardia				
Reported	8			
Not specified	7	0.82	0.070	-7.22
Proportion of seizures with IT				
<82.5 % (median)	5			
≥82.5%	6	0.004	0.023	58.92
Not reported	4			
Time of onset of pre-ictal HRI (seconds prior to seizure onset)				
≤5 sec (median)	4			
>5 sec	3	0.039	0.017	55.54
Not reported	8			